



BANGALORE INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to VTU, Belagavi)

K. R. Road, V. V. Puram, Bengaluru – 560004

ASSIGNMENT – 1

“Personal Portfolio Website Development”

Submitted to:

Department of Electronics and Communication

Subject Code: 1BESC204E

Faculty Name: Gahan A V

Submitted by:

Name: R kruthik raj

USN: 1BI25EC108

Roll Number:47

Section: EC-2

Date of Submission: 08-05-202

1. Introduction

A personal portfolio website is a digital platform used to present personal information, educational background, technical skills, achievements, projects, and contact details in an organized and visually attractive manner.

This project focuses on developing a futuristic cyber-themed personal portfolio website using HTML, CSS, and JavaScript. The website is designed with advanced animations, glowing effects, responsive layouts, and interactive particle backgrounds to create a modern user experience.

The website represents the personal profile of R Kruthik Raj, a Bachelor of Engineering student from Bangalore Institute of Technology.

2. Objectives of the Project

The main objectives of this project are:

To create a professional online portfolio.

To present personal information in a creative way.

To improve web development skills.

To design an attractive and responsive website.

To implement animations and interactive effects.

To showcase technical skills and achievements.

3. Technologies Used

Technology	Purpose
HTML	Structure of the website
CSS	Styling and animations
JavaScript	Interactive functionality
Particles.js	Animated particle background
Google Fonts	Futuristic typography

4. Software Requirements

The following software tools are required for this project:

Visual Studio Code / Any Text Editor

Web Browser (Chrome, Edge, Firefox)

Internet Connection (for fonts and particles.js library)

5. Hardware Requirements

Minimum hardware requirements:

Processor: Intel i3 or above

RAM: 4 GB or above

Storage: 1 GB free space

Display: 1366 × 768 resolution or higher

6. Website Features

The portfolio website contains the following features:

6.1 Navigation Bar

Sticky navigation bar

Smooth scrolling effect

Hover glow effects

6.2 Hero Section

Large animated heading

Introduction tagline

Interactive button

6.3 About Section

Personal information

Educational details

Profile photograph

6.4 Skills Section

Technical and non-technical skills

Hover animation effects

6.5 Achievements Section

Workshops

Mini projects

Hackathon participation

6.6 Interests and Hobbies

Drawing

Gaming

Problem solving

Exploring technology

.

6.7 Contact Section

Email

Phone number

LinkedIn profile

GitHub profile

6.8 Particle Background Animation

Interactive animated particles

Hover interaction

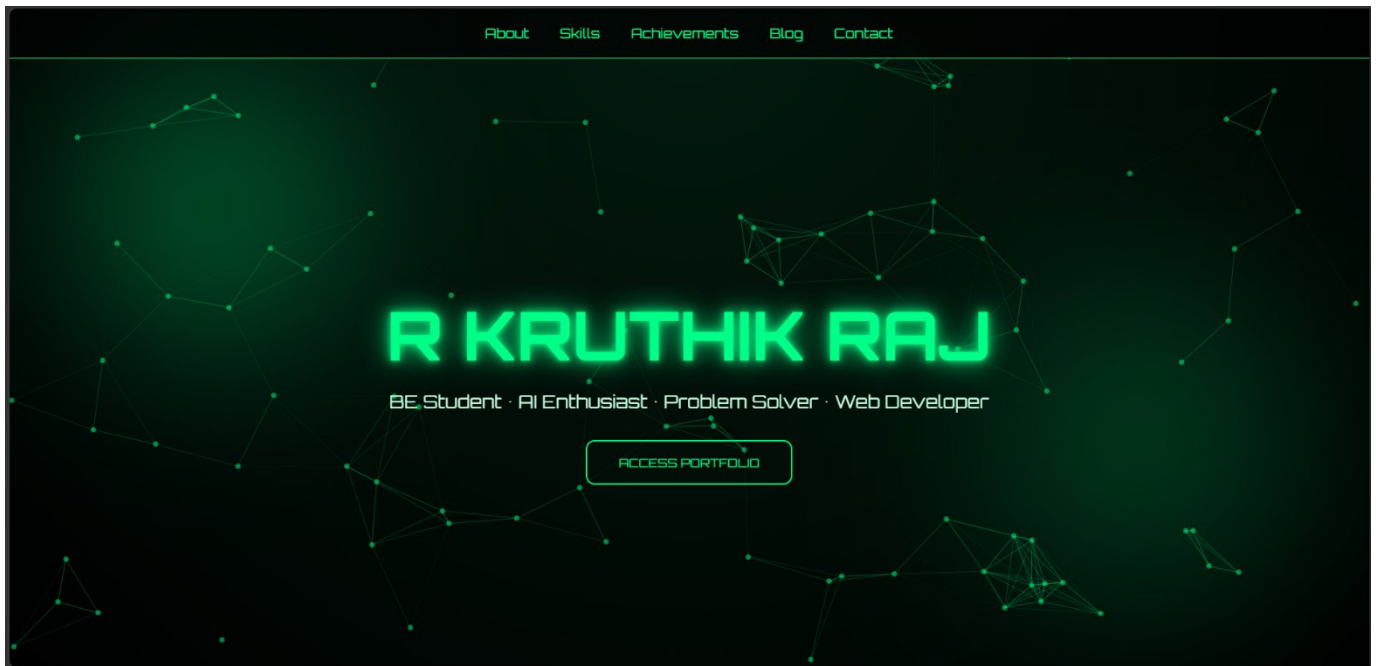
Modern cyber appearance

7. screenshots of the website

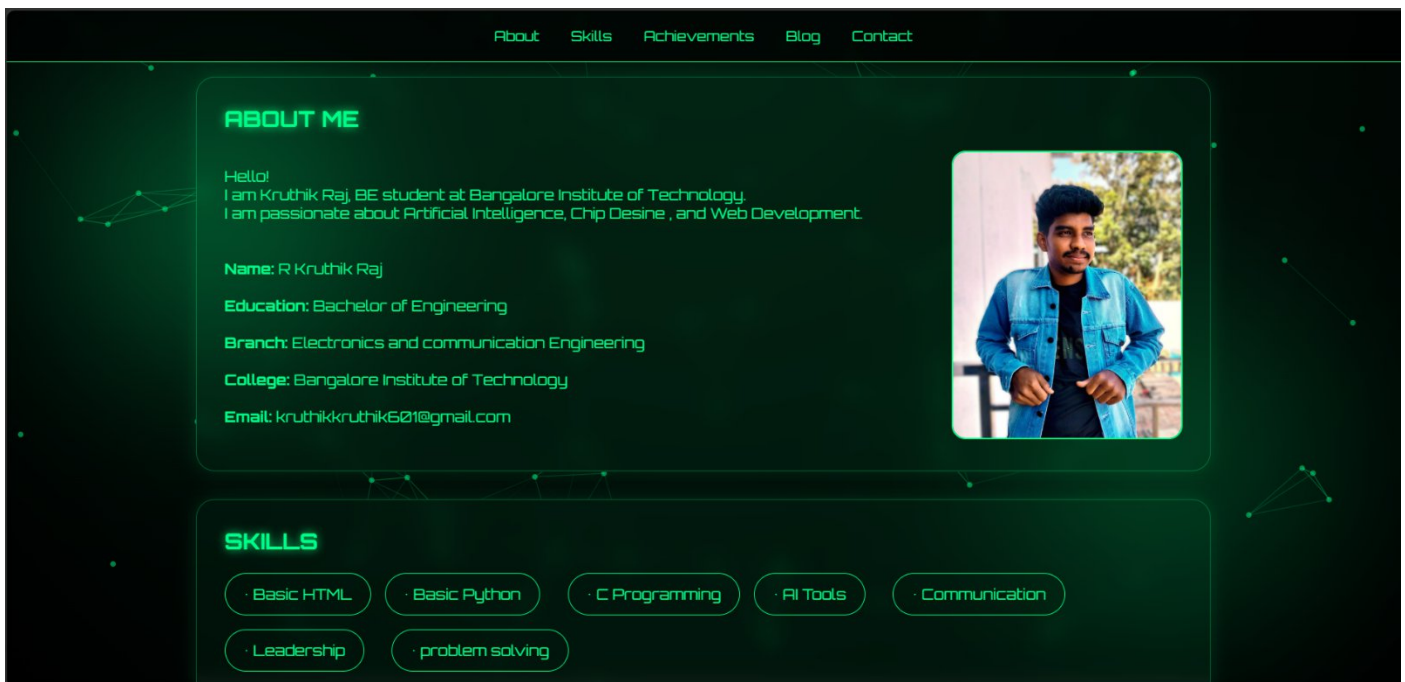
The following placeholders represent each major section of the website. Screenshots should be taken by opening the HTML file in Google Chrome and capturing each section using the browser's built-in screenshot tool or a screen capture application.

The screenshots are attached in the next page

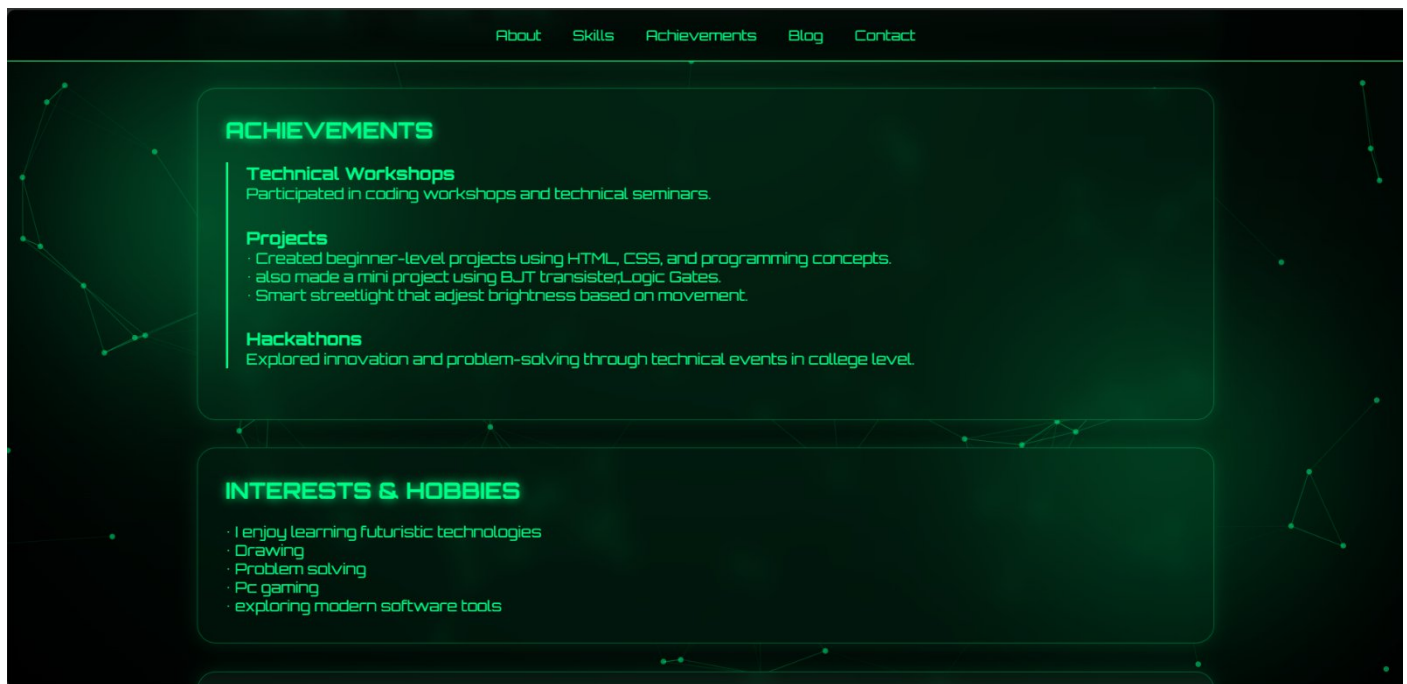
7.1 Hero \ Home section:



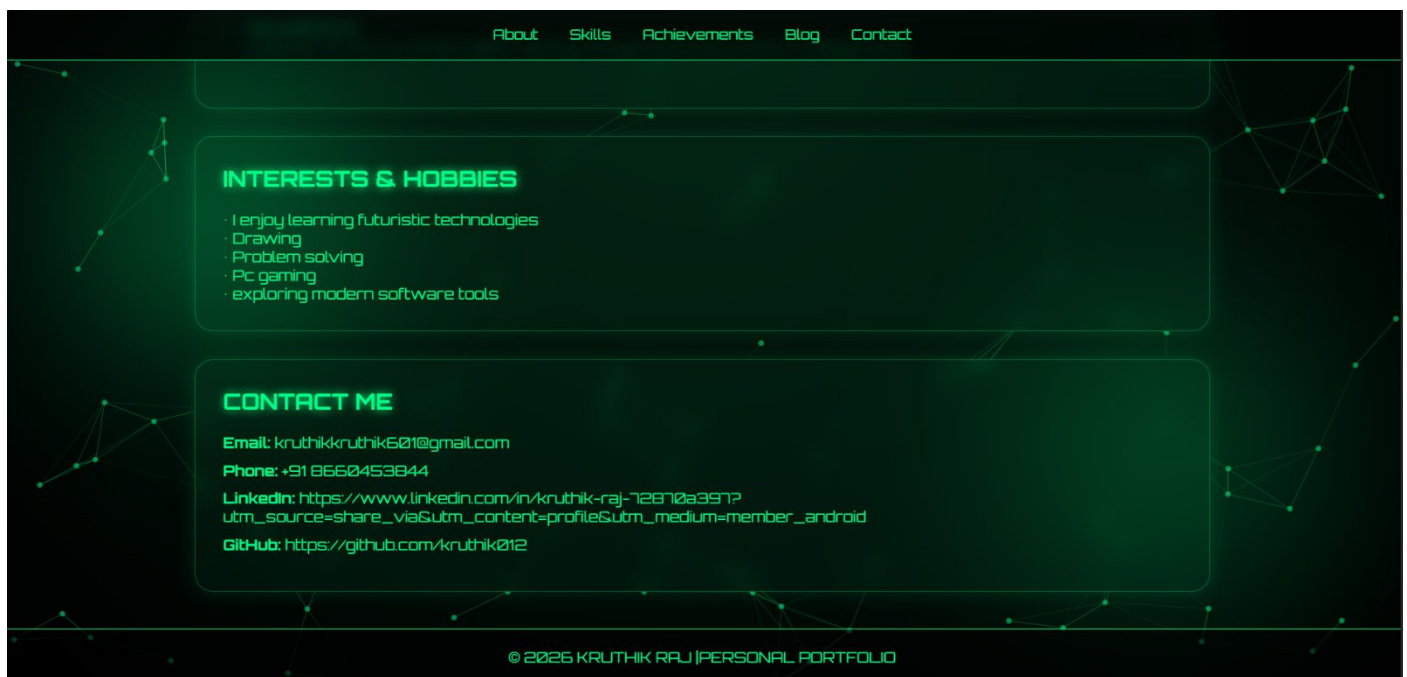
7.2 About me and skills section:



7.3 Achievements and Hobbies section:



7.4 Interest and contact section:



8. System Design

8.1 Website Structure

The website is divided into multiple sections:

Header Section

Navigation Section

About Section

Skills Section

Achievements Section

Interests Section

Contact Section

Footer Section

9. Explanation of Code

9.1 HTML

HTML is used to create the structure of the website.

Examples:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Kruthik Raj Portfolio</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div id="particles-js"></div>
```

```
<nav>
```

```
<a href="#about">About</a>
```

```
<a href="#skills">Skills</a>
```

```
<a href="#contact">Contact</a>
```

</nav>

<header>

<h1>R KRUTHIK RAJ</h1>

<p>BE Student • AI Enthusiast • Web Developer</p>

</header>

<section id="about">

<h2>About Me</h2>

<p>

I am Kruthik Raj, a BE student at Bangalore Institute of Technology.

Interested in AI, Chip Design, and Web Development.

</p>

</section>

<section id="skills">

<h2>Skills</h2>

<div class="skills">

HTML

Python

C Programming

AI Tools

</div>

</section>

<section id="contact">

<h2>Contact</h2>

<p>Email: kruthikkruthik601@gmail.com</p>

<p>GitHub: github.com/kruthik012</p>

</section>

<script src="https://cdn.jsdelivr.net/particles.js/2.0.0/particles.min.js"></script>

<script src="script.js"></script>

</body>

</html>

9.2 CSS

CSS is used for:

Styling

Layout design

Animations

Glow effects

Responsive design

Example:

```
body{  
    margin:0;  
    font-family:Arial;  
    background:black;  
    color:#00ff88;  
    text-align:center;  
}  
#particles-js{  
    position:fixed;  
    width:100%;  
    height:100%;  
    z-index:-1;  
    background:black;  
}  
nav{  
    padding:20px;  
}  
nav a{  
    color:#00ff88;  
    margin:15px;  
    text-decoration:none;  
}  
header{  
    padding:120px 20px;
```

```
}  
section{  
    padding:40px 20px;  
}  
  
.skills span{  
    border:1px solid #00ff88;  
    padding:10px 15px;  
    margin:10px;  
    display:inline-block;  
}
```

9.3 JavaScript

JavaScript is used for interactive particle animation.

Example:

```
particlesJS("particles-js", {  
    particles: {  
        number: {  
            value: 80  
        },  
        color: {  
            value: "#00ff88"  
        },  
        line_linked: {  
            enable: true,  
            color: "#00ff88"  
        },  
        move: {  
            enable: true,  
            speed: 2  
        }  
    }  
});
```

10. Responsive Design

The website is responsive and works properly on:

Desktop computers

Tablets

Mobile phones

Media queries are used to adjust layouts for smaller screens.

Example:

```
@media(max-width:768px){  
  .hero h1 {  
    font-size:45px;  
  }  
}
```

11. Output Description

The final output is a futuristic cyber-themed portfolio website with:

Animated particle background

Interactive cards

Smooth scrolling navigation

Responsive layout

Personal and professional details

The website provides a modern and visually impressive user experience.

12. Conclusion

The Personal Portfolio Website project successfully demonstrates the use of HTML, CSS, and JavaScript for creating an interactive and responsive web application.

The project improves web development skills while also creating a professional digital identity. The futuristic cyber theme, glowing effects, and particle animations make the website unique and visually appealing.

This project can be further expanded with additional features and advanced technologies in the future.